Nishal Stanislaus Silva, PhD
Toronto, ON, Canada
nishal.silva@hotmail.com | +1 613 200 2030
nishal.xyz | linkedin.com/in/nishal-silva | github.com/ns2max

Dear Hiring Manager,

My PhD thesis is, in essence, hardware-aware ML: designing a model to operate correctly and quickly under a fixed latency, compute, and power envelope on embedded ARM hardware, then proving it under load. That work culminated in a real-time audio pattern detector running at 14 ms inference on a Raspberry Pi 4, with F1 0.76 at 31.4% CPU utilization – 74x faster than the 1038 ms DTW baseline it replaced. This is the exact discipline your Senior Researcher, Edge AI Optimization / Hardware-Aware ML role calls for, and it is the discipline I have spent the last five years practicing on real embedded devices rather than in simulation.

That discipline runs deeper than a single result. As a postdoctoral researcher at the University of Trento, I designed frozen-protocol benchmark suites and ablation studies – RNN versus DTW, audio versus MIDI representations, pattern-set sizes of 1, 3, and 10 – that quantify precisely where accuracy, latency, and CPU usage trade off against each other, and I kept a written record of the negative results as well as the positive ones. I also built MIRaaS, a UDP edge-to-server offload architecture accepted at IEEE IS2 2026, which characterizes the latency and compute trade-offs behind the decision of what should run on-edge versus what should be offloaded – directly relevant to any team optimizing where inference happens. Earlier, as a visiting researcher at McGill University's IDMIL/CIRMMT lab, I built a self-powered smart electric guitar and profiled its compute and power draw to validate real-time feasibility on hardware with no external power supply – power-budget engineering in the most literal sense.

I should be direct about one gap: my hardware-aware optimization experience is with audio and time-series models, not large-scale foundation-model compression or distillation. I have not compressed a transformer-scale language model. What I bring instead is a methodology – profiling on real hardware, frozen-protocol benchmarking, and quantified accuracy/latency/CPU trade-off analysis – that is domain-general and transfers directly to any model class once the target hardware and constraints are defined.

I am a Canadian Permanent Resident and do not require sponsorship to work in Canada. I am available immediately and open to relocating to Edmonton for this role. I would welcome the chance to discuss how this background applies to your team's work.

Sincerely,
Nishal Stanislaus Silva